CPT 314 CROPPING SYSTEM

ALLEY CROPPING

Alley cropping is the cultivation of food, forage, or specialty crops between rows of trees. It is a arger version of intercropping or companion planting conducted over a longer time scale.

Or, Alley cropping is a farming system in which arable crops are grown in alleys formed by trees or shrubs established mainly to hasten soil fertility restoration and enhance soil productivity.

Or, Alley cropping is a type of agroforestry where crops are grown in rows between strips of perennial trees or shrubs. This type of farming is thought to improve soil health and provide complementary benefits to the crops, such as shade, nitrogen fixation, and pest control. Alley cropping has been shown to be an effective way to increase crop yields and improve soil quality in tropical environments.

Or, Alley cropping is a sustainable agriculture technique that involves growing crops between rows of trees. This technique can improve soil health, provide shade and wind protection for the crops, and help to reduce erosion. Alley cropping is a complex agricultural technique that has many benefits for farmers and the environment.

Why practice alley cropping?

Alley cropping can have many ecological and economical benefits.

- **Reduced soil erosion.** Especially when established in sloping areas, alley cropping can slow erosion, thereby improving water quality.
- **Improved crop performance.** Added organic matter from trees and shrubs can increase soil productivity. Alley cropping also creates a microclimate from the increased shade and reduced wind, which in turn increases water use efficiency by crops.
- **Reduced use of chemical fertilizers and pesticides.** Weeds are better controlled, and increased nutrient flow improves soil fertility without the use of fertilizers.
- **Minimized nitrogen leaching.** Nitrogen leached beyond the cropping root zone is often captured by the deeper tree root systems. Minimizing this leaching improves water quality.
- **Promotion of biodiversity.** Planting diverse vegetation provides more wildlife with more potential habitat.

Design considerations

When designing an alley cropping system, a landowner must consider a few things:

- Amount of rainfall.
- Compatibility of trees and shrubs with crops to minimize competition for water, nutrients, and light.
- Spacing between and within rows.
- Sun direction.
- Maintenance requirements and available equipment.

Establishing an alley cropping system

Alley cropping systems can be designed and established depending on the landowner objectives. There are several ways of establishing alley cropping systems:

Convert existing orchard into an alley cropping system by integrating agronomic or forage crops. Thinning of the orchard maybe employed to allow for the establishment of alley cropping systems.

Cultivate forage crops between rows of trees or shrubs, which can be used for cattle and livestock.

Alternative perennial crops such as medicinal plants or Christmas trees within tall field crops such as corn.

Spatial arrangement
Alley cropping is an agroforestry practice where crops are grown between lines of trees and / or shrubs that are managed and spaced at regular intervals in cropland. This practice has received much research attention and is regarded as having promise for solving problems of declining soil fertility in situations where farmers cannot afford to use inorganic fertilizers at the recommended rates.

Areas where the practice is relevant
The practice of alley cropping is not yet widespread among farmers in Kenya. Research findings indicate that alley cropping is not feasible where average rainfall is less than 800 mm annually. The practice has its major potential in humid lowlands, e.g. in the coastal strip. Since the technology is labour demanding, it is relevant mainly in areas with small farms and a high population density or, in other words, where labour is not a limiting factor.
Establishment and spacing
The establishment of hedgerows requires many trees or shrubs and therefore a cheap propagation method is called for. Direct seeding or use of cuttings would be ideal, but so far seedlings have been used in most research experiments. If seedlings are to be raised, on-farm nurseries are recommended since growing the relevant species does not require

much skill. Calliandra may be an exception due to the scarcity of seed supplies in Kenya, and expensive seeds are often imported.
The spacing used in field trials has ranged from 4 to 8 m between rows and from 0.25 to 2.00 m within rows. In humid areas, close spacing can be tolerated, but in drier conditions a wider spacing is required if competition for moisture is not to be too severe. On flat land, hedgerows should be oriented in an east-west direction to reduce shading. On sloping land, hedgerows must be oriented along the contours.

Designs Unique to Each Landowner
Each alley cropping practice can be uniquely designed to reflect landowner needs and site potential. However, there are physical interactions between the rows of woody species and the companion crop that should be understood and reflected in plans for the alley cropping practice design. Physical growth traits of individual trees and how these influence the crop, are based on three factors:
A. Light Competition
B. Root Competition for Water and Nutrients
C. Allelopathy

A) Light Competition (above ground)
A tree species should be selected that best accommodates the sunlight requirements of a specific crop. Some tree species have small leaves and feathery foliage casting a light shade and lend themselves well to alley cropping. When considering a tree species for an alley cropping practice, small leaves and light shade is preferable to heavy shade. In what ways can light competition be reduced?
• Spacing
When the distance between rows of trees/ shrubs is increased, the years an alleyway may be cropped with minimal competition from the trees is also increased.
• Row Orientation
An East-West orientation of tree rows will maximize the sunlight received by an alley crop, provided the topography permits this arrangement. Trees may have to be planted on the contour if erosion is a consideration. Trees may be planted in other orientations if prevailing winds have a negative influence on crop yields.
• Maximizing available light
Trees with small fine leaves will allow morelight through the canopy. These leaves decompose rapidly and allow nutrients to be recycled into the soil faster. As decomposed materials contribute organic matter to the soil, the soil moisture-holding capacity is increased. Increased organic matter enhances soil microbe and earthworm activity. This also improves soil tilth and health.
• Understanding crown and foliage characteristics - phenology
Utilize trees that leaf-out late in the spring and/ or drop leaves early in the fall. If the agronomic crop component matures in the early spring, such as winter wheat (Triticum aestivum L.), a tree that breaks dormancy late is ideal; alternatively, a crop that is harvested late in the fall is appropriate with a tree that drops its leaves early. Black walnut (Juglans nigra) is a good example of a tree that satisfies both scenarios.
• Timely Thinning and Pruning
Properly thinning trees within rows can maintain semi-open crown conditions.

Maintained through regular thin
growth of shade intolerant comp
reach 1" in diameter improves f
permitting more sunlight to reac

B) Root Competition (below groun
Competition for water and nutri
only affects the yields of the cor
ways can root competition be re
• Understanding rooting zones
Vertical distribution of root syst
a reduced volume of roots near t
adjacent crops). Erosion can be
management.
• Which trees have deep roots?
Tree species have different site r
tend to be shallow rooted. While
appendices, it is always useful to
determining species suitable to y
• What can be done if the trees a
Through the use of a Ripper, Co
addition of sub- surface knives r
Trenching trials strongly sugges
competition for water and/or nut
With early (beginning with your
lateral roots, the number of tree
The remaining roots will remain

C) Allelopathy (Chemical Interacti
Traditionally, the term allelopath
by one plant on the growth of ne
acids that inhibit growth of plan
produce a compound called jugl
A broader definition would also
plants (e.g., legumes, European
nearby plants. Chemical interact
combinations that work together
Conversely, nitrogen-fixing plan
extra nitrogen.

Tree arrangement
The tree and/or shrub row(s) are pla
the purpose, either on the contour or
troublesome winds. Several factors
of trees or shrubs including slope le
equipment width.

TAUNGYA SYSTEM OF FARMING

The Taungya System is a form of agroforestry system in which short-term crops are grown in the early years of the plantation of a woody perennial species to utilize the land, control weeds, reduce establishment costs, generate early income, and stimulate the development of the woody perennial species.

This is a modified term for "shifting cultivation," in which labor is permitted to raise a crop in an area but only side by side with the forest species planted by them. The practices consist of land preparation, tree planting, growing agricultural crops for 1–3 years until shade becomes dense, and repeating the cycle in a different area. Traditional Taungya consists of land preparation for tree plantations, growing agricultural crops for 1–3 years after the tree plantations, and moving on to another location to repeat the cycle.

Types of Taungya

There are three types of Taungya

1. Departmental Taungya: Under this, agricultural crops and plantations are raised by the forest department. By employing daily paid laborers, raising agricultural crops and the plantation keeps the land free of unwanted vegetation.

2. Leased Taungya: The forest land is given on lease to the person who affects the highest money for raising Agril. Crop for a specialized number of years and ensure the care of tree plantation.

3. Village Taungya: This is the most successful of all the three Taungya systems. Under this, the people who have settled down in a village inside the forest raise crops. Usually, each family has about 0.8 to1.7 ha of land, raising trees and cultivating crops for 3 to 5 years.

Advantages of Taungya System

1. Artificially, regeneration of the forest is done at a cheaper rate.
2. Problem of unemployment is solved to some extent.
3. Better utilization of land.
4. Remunerative to the forest department.

Disadvantages of Taungya System

1. Creates some legal issues.
2. Exploitation of human labour.
3. Danger of epidemic.
4. Loss in soil fertility.

How To Make Taungya System (Procedure)

The Taungya system begins with the harvesting of the forest. The harvested area is clear-cut and burned. Some trees are reserved for shade or fruit. Usually, the site selected for the Taungya system should be flat and fertile. Generally, one or two hectares of land are allocated to one family. However, sometimes the area may be larger.

Usually, cultivation of an agricultural crop is permitted for one or two years before forest plantation. This helps the farmer take one or two good harvests in the highest fertility condition of forest land. Allowing one or two crops before plantation is an incentive to the farmers for clearing the land for forest plantation.

When tree crops are planted with agricultural produce in the second or third year, they receive the intensive cultural operation benefits the farmer adopted for his agricultural crops. When the forest plantation gets established, the farmers have to leave the area.

CROPPING SYSTEM

A cropping system is defined as the cropping pattern and its management to derive benefits from a given resource base under a specific environmental condition. The productive base of cropping system is plant growth, which is influenced by management and environment. Management here includes all those components associated with crop production. These include, crop area, crop biomass, economic yield, crop rotation, crop calendar, time and spread of sowing and harvest. Environment is the invariant resource of soil, physiography and climate.

Cropping systems are grouped into:

a. Monocropping
b. Mixed cropping
c. Continuous cropping
d. Multiple cropping
e. Crop rotation.

a.monocropping

Monocropping is the agricultural practice of growing a single crop year after year on the same land, in the absence of rotation through other crops or growing multiple crops on the same land, which is known as polyculture. When it comes to popular crops used for monocropping, corn, soybeans, and wheat are three common crops often grown using monocropping techniques. he method of monocropping allows for farmers to have consistent crops throughout their entire farm. Then the farmers plant their most profitable crop only, using the same seed, pest control, machinery, and growing method on their entire farm, which may increase overall farm

profitability. Those in favor of monocropping claim it is a more profitable way to farm than switching crops around each year. While this may be true in the short term, the effects of monocropping can be devastating for the environment.

Advantages of monocropping

1. It makes it possible the use of machines in farm operation.

2. Mono-cropping leads higher productivity per hectare,

3. It also leads to specialization among farmers.

4. The control of weeds is easy. This is because herbicides can be used

5. It employs more labor thereby leading to greater harvest

Disadvantages of mono-cropping system of cropping

1. It is risky because crop failure arising from pest, diseases or weather conditions will result in total loss of income to the farmer for that year.

2. The system encourages the rapid spread of pests and diseases on the farm.

3. Labour may not be efficiently utilized throughout the year.

4. It does not afford the farmer a variety of crops. types of cropping system

5. Most often it requires high capital investment.

6. It can't be practiced within an area where there is community farming

b.Mixed cropping types of cropping systems

This is also called multiple cropping because it involves the planting of more than one type of crop on the same farmland in one planting season

. It is very common under subsistence agriculture and in are where farmlands are limited. Under mixed cropping, the farmer could practice any of the following:

Inter-planting types of cropping systems

: This is the growing of two crops together on the same land. The crop which was planted first is also harvest first while the one planted last remains on the plot to harvested later.

An example is the growing of maize, cassava, groundnut, melon, yam and okra together on the same piece of land in one planting season. Maize, which is usually planted first, is also harvest first. Maize is therefore said to be inter-planted with yam.

(b) Inter-cropping: This is when two crops are grown together with the crop planted last being harvested first. Usually the c planted last has shorter lifespan than the one planted first, example is the planting of melon after yam has been plant The melon will be harvested first while the yam continues on plot. Yam is therefore said to be intercropped with melon. types of cropping system

Advantages

1. It affords the farmer a variety of crops.

2. It serves as insurance against the failure of one type of crops.

3. It minimizes the spread of diseases and pests on the farm.

4. It enables the crops to make efficient use of soil nutrients.

5. The ensures efficient utilization of labour throughout the year.

Disadvantages

1. It does not encourage the use of machines on the farm.

2. It may lead to rapid exhaustion of soil nutrients if legumes not included.

3. It is labour intensive.

4. Pests and disease of crops agents may persist on the farmland. This is because there are always food and alternative hosts for them.

c. Continuous cropping types of cropping system

This is the practice of putting a farmland under cultivation continuously that is from year to year. It may take any of form: this type of farming practice usually lead to soil loss of nutrients, becoming acidic

(a) Annual cropping:

Planting annual crops which are replaced after harvesting. This means the land is cleared, tilled and cropped every season. read more on pre-planting operations here . This is common where land is scarce. this type of farming is mostly practiced in the southern parts of Nigeria. types of cropping system

(b) Permanent cropping:

This involves planting and maintaining the crops, usually permanent crops continuously on the farm land, it usually in plantations. read cultural practices in crop production

Advantages

1 It reduces the cost of land preparation after the initial clearing and tilling.

2 It enables the farmer to construct permanent structures such as storage structures on the farm.

3 It can be practiced where land is scarce.

Disadvantages

1. The fertility of the soil is easily exhausted.

2. It leads to destruction of soil structure.

3. It encourages soil erosion.

4. Yields me normally reduced with increasing years of cropping.

5. It encourages build-up of crop pests and disease agents.

6. It required high amount of money to keep the land fertile and productive.

d. Crop Rotation type of cropping systems

This involves the planting of different types of crop in different plots on a farmland during one season; and at the beginning of the next season, the crops are changed from their respective plots, while following a definite order or sequence. types of cropping system

The system combines mixed cropping with continuous cropping and is mainly practiced by institutions of learning.

For crop rotation to be successful, certain principles must be followed

Principles of Crop Rotation

(a) The same type of crop should not be allowed to follow each other on the same plot. For example, maize should not follow maize.

(b) Crops that belong to the same group should not also follow each other on the same plot, e.g. cassava should not follow yam, or to follow maize.

(c) Crops that have deep roots like yam and cassava, read more about classification of crops here.. should be followed with those that have shallow roots such as maize and groundnut.

(d) Crops that consume a lot of nitrogen such as the-cereal group should be followed by those that add nitrogen to the soil such as maize and the legume group,

(e) Crops likely to be affected by the same disease or pest should not follow each other on the same plot. The number of crops involved in the rotation will determine the type of rotation. Therefore, there could be a two-year, three-year, or tour-year crop rotation.

How to Design a Four-Year Crop Rotation

(a) Divide the farmland into four plots.

(b) Choose the crops to cultivate.

(c) Plant one crop on each plot, making sure the principles guiding the adoption of the system are adhered to.

(4) At the end of one season, shift the crop from plot B to A, C to B, D to C and A to D

e. Multiple cropping

Definition: It is the practice of growing two or more two crops on the same field in a year. Multiple cropping involves the intensification of cropping in temporal and spatial dimensions. Multiple cropping is a philosophy of maximum crop production per land area with a minimum of soil health deterioration.

Multiple cropping philosophy

The philosophy is based on the concept that:

- The high output is good for the soil;
- The minimum tillage promotes soil tilth and conserves soil organic matter, thus resulting in good physical conditions;
- The cover in the form of living mulch is a protection against all forms of erosion and weeds;
- Residues of manures and fertilizers utilize appropriately;
- The successive crop provides an opportunity for the efficient utilization of land, labor, capital, and other available resources;
- There is scope for higher production with a higher rate of outturn.

INTERCROPPING

The intensification of monocultures has led to the degradation of soil systems worldwide, but the emergence of intercropping practices may provide a key alternative to restabilizing sustainable food production. Intercropping, therefore, capitalizes on the mutualistic and beneficial relationships among different crops, harnessing the ecological processes of niche differentiation to ensure the higher quality and sustainability of crop production.

Previous studies have found this practice increases productivity and provides greater ecosystem services, and can be applied across many environmental conditions, soil types, and crop combinations.

Types and functions of intercropping systems

To date, intercropping has been further subdivided into four main types that offer similar but varying benefits depending on crop types, environment, and objectives.

First, **mixed intercropping** consists of the proximate and simultaneous growth of two or more crop types without spatial arrangement, allowing for crops to be seeded and harvested together and separated after harvest.

Second, **strip intercropping** is a system of parallel strips of crops each made of rows of different crops. Strips are therefore spatially arranged and need to allow for harvesting space as well as inter-crop interactions.

Third, similar to strip cropping, **row intercropping** differs only as one of the crop types are planted in single or double rows beside the other, allowing for higher interspecific interaction. This can promote shading, root mingling, and competition for water and nutrients. This strategy in particular is gaining traction due to its beneficial effects across crops including disease suppression.

Finally, **relay intercropping** combines strip and intercropping planting arrangements by cultivating two or more different species simultaneously but crops are planted and harvested at different times. This system also allows for spatial arrangements in strips or rows, and is unique due to the opportunity for coordination between crop life cycles, allows for isolated growth before crop interaction, and also provides a recovery period for the second crop type once the first is harvested.

Each strategy emphasizes different elements of intercropping, with varying levels of crop interaction, but challenges are also key to consider. For instance, spatial allocation of strip intercropping or the fact that relay cropping requires an advanced understanding of the effectiveness in planting different crop types, present important technical management considerations.

Advantages of inter-cropping

- In a carefully planned inter-cropping the LER is usually greater than one. This means that there is yield advantage in growing crops together than growing each one separately. This advantage may rise from several sources.
- The crops may complement one another in their use of field time. The period of their peak demands for light, water, nutrients and other resources may differ, so that in general there is a more efficient utilization of the resources available. For example, during the two or three month that yam takes to sprout and establish adequately on the field, a quick inter-crop of maize or melon would beneficially utilize the field resources during this period.
- The component crop may complement each other in their use of space. For example, an inter-crop of a deep rooted crop can exploit various horizons of the soil.
- An inter-crops may be able to utilized resources which the main crop may not be able to utilized or which may even be disadvantageous to it.
- Certain crops may exert specific beneficial effect on others. For example, plantains inter-cropped with young cocoa seedlings provide shade for the seedlings. Similarly, in an inter-crop of a legume with a cereal crop, the cereal would benefit from the nitrogen fixed by the legumes.
- By having many crops growing simultaneously on the field the farmers is more or less buffered against failure of one of the crop.
- Inter-cropping allows for a more uniform distribution of labour throughout the year.
- When one component of an inter-crop combination fails, the other combinations are able to utilized the resources that would have been available to the failed crop, and so yield better than they would have done otherwise. In other words there is yield stability.

- The spread of diseases and pests is less rapid than in sole cropping. This is probably because the mean distances between the plants of the same component crops are greater. In many instances, the other component crops are not susceptible to the particular disease or pest afflicting one component and may act as physical barriers to the spread of diseases and pest

Disadvantages of inter-cropping

- Since many crops exist together on the field, it is not possible to tailor production practices to the needs of any particular crop.
- Control of pests and diseases is particularly difficult because pesticides which have been development to control a disease on one particular component crop may have deleterious effect on other crops in the combination.
- It is difficult to mechanize operations such as planting, weeding and harvesting.

Evaluating yield from inter-cropping

The relative yield of each component crop in an inter-cropping situation is the yield of that component in the inter-cropping situation divided by what that crop would have yielded as a sole crop, covering the same area as the inter-crop and managed at the same level.

Suppose, for example, that a field with a crop combination of maize and cow-peas yields 1.5 tonnes/hectare of maize and 0.25 tonnes/ha. Of cow-peas. If the expected sole crop yield of maize is 2.0 t/ha and that of cow-peas is 0.5 t/ha. Then the relative yield of maize is 1.5/2.0 =0.75 and the relative yield of cow-peas is 0.25/0.50=0.50.

The sum of the relative yields of the various component crops in the inter-crops in the inter-crop is sometimes called the relative yield total. A little reflection of how many times the land area used for inter-crop would be required to produced the same yields of the component crops when they are grown as sole crops. The relative yield total is therefore more conventionally referred to as **land equivalent ratio (LER).**

SHIFTING CULTIVATION

Shifting cultivation is a mode of farming long followed in the humid tropics of Sub-Saharan Africa, Southeast Asia, and South America. In the practice of "slash and burn", farmers would cut the native vegetation and burn it, then plant crops in the exposed, ash-fertilized soil for two or three seasons in succession. As the original organic matter reserve in the topsoil decomposed and as the high rainfall would leach out the nutrients from the root zone, the farmers would abandon the cleared plot and move to an adjacent patch of forest. They would allow each cultivated plot to recover its vegetation and fertility for some time. At about fifteen or twenty years before returning to it. Thus, they practiced an extensive rotation (forest-crop-forest) that was sustainable for many generations, while the population density remained low. What disrupted the system was the progressive growth of population that has taken place in the last century. Population pressure has forced farmers to return to the same plots earlier before the soil had been given the time to be completely rejuvenated. Soil fertility then began to deteriorate, owing to the extraction of nutrients without replenishment and to progressive erosion of the bared soil.

Shifting cultivation and climate

Shifting cultivation is mainly practiced in the humid tropical areas of sub-Saharan Africa, Southeast Asia, Central America and South America. In these regions, the average monthly temperature is more than 18°C year-round and the growing period is characterized by 24-hour average temperatures greater than 20°C. Further, the growing period extends to more than 180 days.

In addition, these areas typically have high levels of rainfall and year-round humidity. The rainfall in the Amazon basin in South America is more or less consistent throughout the year. In sub-Saharan Africa, however, there is a distinct dry season with 1-2 months of low rainfall.

Shifting cultivation and climate change

Burning biomass to clear the land in this agrosystem results in releasing carbon dioxide and other gases into the atmosphere. If the shifting cultivation system is in equilibrium, the released carbon dioxide should be re-absorbed by the regenerated vegetation when the land is left fallow. Unfortunately, the system is not usually in equilibrium because of either the shortening of the fallow period or the utilization of the plot for another type of land use instead of leaving it in fallow, among other reasons. Therefore, the net emission of carbon dioxide contributes to global warming and ultimately climate change.

Some researchers have argued that the above scenario is not necessarily true and that shifting cultivation does not contribute to global warming. In fact, it has been posited that these systems are excellent at carbon sequestration. Therefore less carbon dioxide is being released into the atmosphere compared to plantation agriculture, permanent planting of seasonal crops or other activities such as logging.

How is Shifting Cultivation Practiced?

It is a way of farming long followed in the humid tropics of southeast Asia, Sub-Saharan Africa, and South America. In Shifting agriculture in India, farmers would cut and burn the native vegetation. Then they sow crops in the ash-fertilized and exposed soil for **2 or 3 seasons in succession**.

Characteristics of Shifting Cultivation

- Fire is used to clear the land for cultivation.
- Shifting cultivation is a dynamic system which adapts to the prevailing circumstances and is modified as time passes.
- In shifting cultivation, there is a high level of diversity in the types of food crops grown. This ensures there is always food throughout the year.
- Shifting cultivators live both in and from the forest; therefore, they usually also practice hunting, fishing and gathering to fulfil their needs.
- The plots utilised in shifting cultivation typically regenerate more easily and quickly than other forest clearings.
- The selection of locations for cultivation is not made on an ad hoc basis, but rather plots are carefully selected.
- In shifting cultivation, there is no individual ownership of plots; however, cultivators have ties to the abandoned areas.

- The abandoned plots remain fallow for extended periods of time
- Human labour is one of the main inputs of shifting cultivation, and the cultivators use elementary farming tools such as hoes or sticks.

Advantages of Shifting Cultivation

- The crops' growth will start fast in this Cultivation, and sometimes it will get ready for harvest early.
- And there is no danger or fear for the crop-destroying animals and flood in It.
- In this farming, the soil bone diseases are also reduced significantly.
- After slash and burn, it is easy to grow crops.

2. Shifting Cultivation Disadvantages

- There is a loss of 22% of the top of the soil, which is full of fertility in Shifting agriculture. Therefore, it causes the loss of economic rate of people.
- As it has raw sewage and oil residue, water pollution can easily occur in it.
- The Shifting cultivation process also restricts the land-use intensity.
- Loss of biodiversity is also an impact of shifting agriculture, and it is uneconomical.

LAND EQUIVALENT RATIO (LER)

Land equivalent ratio (LER) in agriculture refers to relative land required in area under monoculture or sole cropping to give the same yields as under polyculture or intercropping. The LER ratio refers to the ratio between comparative yield of each crop and tree species in agroforestry compared to the yield in a monoculture system of the same crop and tree species over same period of time.intercropping can be used where a producer can obtain more than one yield in the same area. .

Sustainable intensification of agriculture is required to meet future increasing food demand while reducing the agro-ecological footprint . Diversification of agricultural production systems enhances agro-ecological strategies like increasing biodiversity and higher crop protection.

Intercropping as a strategy increases agricultural productivity per unit of land by improving resource capture . Intercropping can be a guide to diversify agro-ecological systems by using leguminous crops and reducing on the use of mineral fertilizers

Criteria for evaluating different intercropping situations are suggested, and the Land Equivalent Ratio (LER) concept is considered for situations where intercropping must be compared with growing each crop alone. The need to use different standardizing pure crop yields in forming LERs is discussed, and a method of calculating an effective LER is proposed to evaluate situations where the yield proportions achieved in intercropping are different from those that might be required by a farmer. The possible importance of effective LERs in indicating the proportions of crops likely to give biggest yield advantages

The land equivalent ration can be calculated

Applications

The land equivalent ratio can be used whenever more than one type of yield can be obtained from the same area. This can be intercropping of annual crops (e.g. sorghum and pigeonpea) or combination of annual and perennial crops

Minimum tillage

Minimum tillage refers to growing a crop using only that tillage needed for placing the seed at proper depth and then covering them.

Thus minimum tillage aimed that reducing tillage to the minimum necessary for ensuring a good seedbed, rapid germination, satisfactory stands and favourable growing condition.

Methods of Minimum tillage are

1. Row zone tillage
2. Plough plant tillage
3. Wheel track planting

Advantages of Minimum Tillage

1. Improves soil conditions due to decomposition of plant residues insitu
2. Improves soil physical condition due to incorporation of organic residues leading to higher water infiltration into the subsoil
3. Higher infiltration is caused by the vegetation present on the soil and channels formed by the decomposition of dead roots
4. Less resistant to root growth due to improved soil structure
5. Lesser surface runoff of rain water
6. Satisfactory crop stand establishment

7. Less soil compaction by the reduced movement of heavy tillage vehicles and less erosion compare to conventional tillage.

Disadvantages of Minimum Tillage

1. Seed germination is lower with minimum tillage
2. In minimum tillage, more nitrogen has to be added as the rate of decomposition of organic matter is slow
3. Nodulation is affected in some leguminous crop i.e. Fieldpea, Beans
4. Sowing operations are difficult with ordinary implements
5. Continuous use of herbicides causes pollution problem and dominance of perennial problematic weeds.

Zero Tillage

Zero tillage referred as no tillage in which the crop is planted in unprepared soil by opening a narrow slot, trench or band only of sufficient width and depth to obtain proper seed coverage.

The weeds are controlled only by chemical means. Only possible when the soil has --> (a) course textured surface, (b) good internal drainage and (c) adequate quantity of residue as mulch Problems of zero tillage are low seed germination, low mineralization and build up of volunteer plants. Commonly used herbicides under zero tillage are paraquate and diquat

It is an extreme form of minimum tillage. Primary tillage is completely avoided and secondary tillage is restricted to seedbed preparation in the row zone only.

- Placement of seed in to soil without soil preparation
- It is the direct sowing of seed in the field without any disturbance to the soil.
- There are different forms of tillage systems which are practiced by the farmers and their levels of crop residues formations are different for each tillage practices.
- It is a way of growing crops or pasture from year to year without disturbing the soil through tillage
- It is also an agricultural technique which increases the organic matter retention and cycling of nutrients and increases the amount of water that infiltrates in to the soil.
- It reduces or eliminate soil erosion in many agricultural regions.
- Help to increases the amount and variety of life in an on the soil including disease- causing organisms and disease suppressing organisms. The most powerful benefit of zero tillage is improvement in soil biological fertility making soil more resilient. Farm operations made much more efficient, particularly improved time of sowing and better traffic ability of farm operations.

Management of zero tillage

Zero tillage requires some different skills in order to do it successfully with any production system, if zero tillage is not done correctly, yield can drop. So, a combined technique is required for the management of zero tillage.

1. Integrated disease and pest management
Nowadays, it is very popularly adopted in western countries. The approach is very simple. Here the ecosystem is not hurt at all. So, farmers adopt this practice to get optimum yield using zero tillage.

2 Crop rotation
With zero tillage, the residue is kept at the soil surface and thus it increases the soil moisture. But this may also bring the disease or pest outbreak to the crop field. So to manage this problem, farmers should use crop rotation, by the rotation of crops, the completion of pathogen life cycle can be stopped and thus they can be easily managed.

3.Use of light implements
The proper zero tillage can be done with the help of light implements. So, it is necessary to use light tractors as it escapes the chances of soil compaction